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PI Name:	Ferl, Robert Ph.D.		
Project Title:	Functional genomics of plant response and adaptation to low atmospheric pressure		
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	This proposal seeks a fundamental understanding of the mechanisms involved in plant responses and adaptations to low pressure environments, with the ultimate goal of producing plants that are specifically designed to thrive in low pressures. The essential drivers of this project are 1) the inescapable engineering limitations to producing orbital, lunar or Martian plant growth facilities that contain earth-normal atmospheric pressures, and 2) the knowledge that plants in certain circumstances can live under reduced pressures. In the initial phases of the project, we propose to use the tools of functional genomics in the model plant system of arabidopsis to examine molecular genetic responses that occur during initial and chronic exposure to atmospheres ranging from near earth normal to near Martian pressures and containing various gas mixtures. Based on the results from these analyses, we will identify genes and metabolic pathways that are activated and repressed in response to low-pressure atmospheres, and predict alterations that might enhance the		

Task Description:	adaptation process. Finally, we will select and directly engineer genetic changes that will test these predictions and potentially lead to plants with enhanced abilities to live in low-pressure environments. The long-term goal of this project is an understanding of the molecular changes in gene activity that accompany the response and adaptation of plants to low atmospheric pressures. Such an understanding will lead to hardware engineering and/or plant genetic engineering modifications that will mitigate the negative impacts of low-pressure atmospheres on plant growth and development, enhance adaptation and strengthen the role of plants as integral and efficient components of extraterrestrial human life support systems. There are four specific aims: 1. determine the spectrum of genes activated and repressed by low pressure 2. develop an understanding of the metabolic pathways involved in response and adaptation 3. determine the effects of defined gas mixtures in mitigating the impacts of low pressure atmospheres 4. develop and test biological engineering strategies to enhance growth in low atmospheric pressures The long-term goal of this project is an understanding of the molecular changes in gene activity that accompany the response and adaptation of plants to low atmospheric pressures. Such an understanding will lead to hardware engineering and/or plant genetic engineering modifications that will mitigate the negative impacts of low-pressure atmospheres on plant growth and development, enhance adaptation and strengthen the role of plants as integral and efficient components of extraterrestrial human life support systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	An understanding of plant responses to extreme environments at the molecular genetic level contibutes to basic molecular biology research. This research will also improve our ability to develop more successful closed system ecologies (green houses etc) on earth as we come to a better understanding of how transpiration, carbon fixation and ethylene metabolism are impacted by reduced atmospheric pressures.
	Conclusions. Arabidopsis plants respond to hypobaria with extensive changes in gene expression patterns. So even though plants have been known to survive exposure to hypobaria, clearly survival in hypobaric conditions requires a dynamic adaptive response. Some of the altered gene expression patterns are similar to those involved in the hypoxic stress response, as would be expected from the low oxygen partial pressures present in hypobaria; yet some changes in gene expression patterns are unique to hypobaric conditions. These unique differential gene expression patterns demonstrate that hypobaria is not equivalent to hypoxia as an abiotic stress and suggest that, for plants, the response to hypobaria is more complex than the acclimation to the reduced partial pressures of oxygen inherent to low atmospheric pressures. By extension, environmental compensation for hypobaric stress in plants is not as simple as increasing the oxygen content as was done for humans in the low pressure environments of the early space age. Hypobaria does not equal hypoxia. Although hypobaria engages pathways distinct from those seen in response to hypoxia, there is significant overlap, indicating that the hypobaria response includes, but is not limited to, a typical response to hypoxia. A relatively small number of genes are induced by both hypobaria and hypoxia , primarily genes involved in fermentative pathways and oxygen transport (Klok et al., 2002). A much larger set of genes is repressed in response both conditions and includes representatives from a wide range of metabolic pathways (see Supporting Information). This wide-spread repression is a well characterized phenomenon of hypoxic stress e.g. (Sachs et al., 1996; Vartapetian and Jackson, 1997; Klok et al., 2002). A much smaller cluster includes genes that were repressed in hypoxic conditions, but were virtually unchanged in hypobaric conditions. The genes representing this group encode several putative proteins that may represent metabolic processes related to oxygen sensing that are mediated through a heme- or iron-based sensor (Quinn et al., 2002). Iron based sensors have been implicated for direct oxygen or indirect redox sensing in various hypoxia-responsive pathways in eukaryotes (Bunn et al., 1998; Zhu and Bunn, 2001; Mecklenburgh et al., 2002), relying in part on subtilisin-catalyzed proteolytic inactivation of p450 that functions in the reduction of Fe(III) in the signal transduction cascade (Aravind and Koonin, 2001). Another small cluster is represented by genes that were induced in hypoxic conditions, and repressed in hypobaric conditions. This group includes additional heme-related proteins, kinases and phosphatases typical of a hypoxic response. The disparity in gene expression patterns displayed in each of these clusters suggests that that hypobaria and hypoxia may differentially activate certain alternative oxygen sensing and transport pathways. Desiccation associated pathways are induced by hypobaria. Cluster A2 defines a large group of genes showing increased expression in hypobaria while being unaffected or repressed in hypoxia, a group largely composed of genes associated with desiccation and ABA signaling related processes (e.g. Ozturk et al., 2002). Examples from cluster A2 include ABI2 (Chak et al., 2000), the APL3 subunit of ADP-glucose pyrophosphorylase (Weber et al., 1995; Rook et al., 2001), LEAs, (e.g. Siddiqui et al., 1998; Kim et al., 2002) dehydrins and rab-like proteins, (e.g. Mantyla et al., 1995; Nylander et al., 2001; Kizis and Pages, 2002) rd29B and cor15B (e.g. Wilhelm and Thomashow, 1993; Yamaguchi-Shinozaki and Shinozaki, 1993), and aldehyde dehydrogenase (Kirch et al., 2001). Thus, it appears that the adaptation response of Arabidopsis to 10 kPa is, at a minimum, a specific adaptation to perceived desiccation which is overlaid upon an adaptation to hypoxia. However, the fact that the plants were grown in a humid environment (>95% rh within the plates), showed no loss of fresh weight or turgor, and yet still responded to 10 kPa as if they were dehydrated, indicates that the desiccation response is likely due to the perception of increased water flux caused by the low pressure environment (Corey et al., 2002) rather than absolute loss of water in the plants. Arabidopsis plants at 10 kPa presented no wilting, browning or other hallmarks of desiccation. This observation begs the question as to whether successful adaptation to a 10 kPa environment actually requires the activation of desiccation related pathways. It is possible that as long as there is sufficient environmental water available in the growth medium, an induction of desiccation-related metabolism may not be necessary for survival. With regard to bioregenerative life support for space, if the desiccation response is not necessary for adaptation to low pressure, then the induction of these pathways could represent a drain on metabolism with a resultant cost to production. Under this scenario, metabolic engineering designed to divert the inappropriate desiccation response could reduce the drain of these pathways on

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metabolism. If, however, the desiccation response is required, then enhancing desiccation tolerance may increase production at low pressure. Given the specificity and detail of the 10 kPa response indicated by the present analyses, studies at intermediate and extended low pressures and mutations in the desiccation and ABA response pathways should clarify the role of desiccation responses to low atmospheric pressures. The present data do suggest that the desiccation stress response is tied as much to the flux of water through stomata as to the physical dehydration of tissue.

Concluding remarks. The experiments and conclusions discussed here expose the need for continued evaluation of the effects of low pressure on biological systems. In some ways, the gene expression data suggest that we actually have only a small fraction of the data necessary to make informed choices on the gas composition and pressures within vessels that support plant growth. By extension, these data also beg the questions of what sort of undetected responses and adaptations to low pressures occur in other organisms, including humans, and whether low pressure environments have contributed to any of the existing spaceflight data on biology and biological responses in spaceflight environments. So the short term questions involve deepening our understanding of responses and adaptation to low pressures for all organisms and biological systems that may be a part of spaceflight and extraterrestrial habitats. Indeed, unpublished molecular data indicate that even the relatively small pressure drops to 75 kPa that occur in the space shuttle effect the expression of hundreds of genes in Arabidopsis. That being the case, even seemingly small choices about operations and procedures conducted within habitat environments can have measurable and dramatic impacts on the biology of those habitats. Therefore very near term decisions involving, for example, the choices of atmospheric composition and pressure in the Crew Exploration Vehicle will have long term implications for biology. In the longer term, the use of low pressure environments may enable life support systems that otherwise would be impossible to construct or maintain under higher atmospheric pressures, keeping viable the concepts of inflatable greenhouses. For example, transparent greenhouses could be erected on Mars using currently available materials, but only if the internal pressure of the greenhouse could be maintained below 7.5 kPa (Boston, 1981). The rather dramatic changes in gene expression that occur at 10 kPa suggest that response and adaptation to lower pressures will be even more extensive, but the study of such pressures are well within the capacity of current experiment chamber design and molecular expression technology.

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